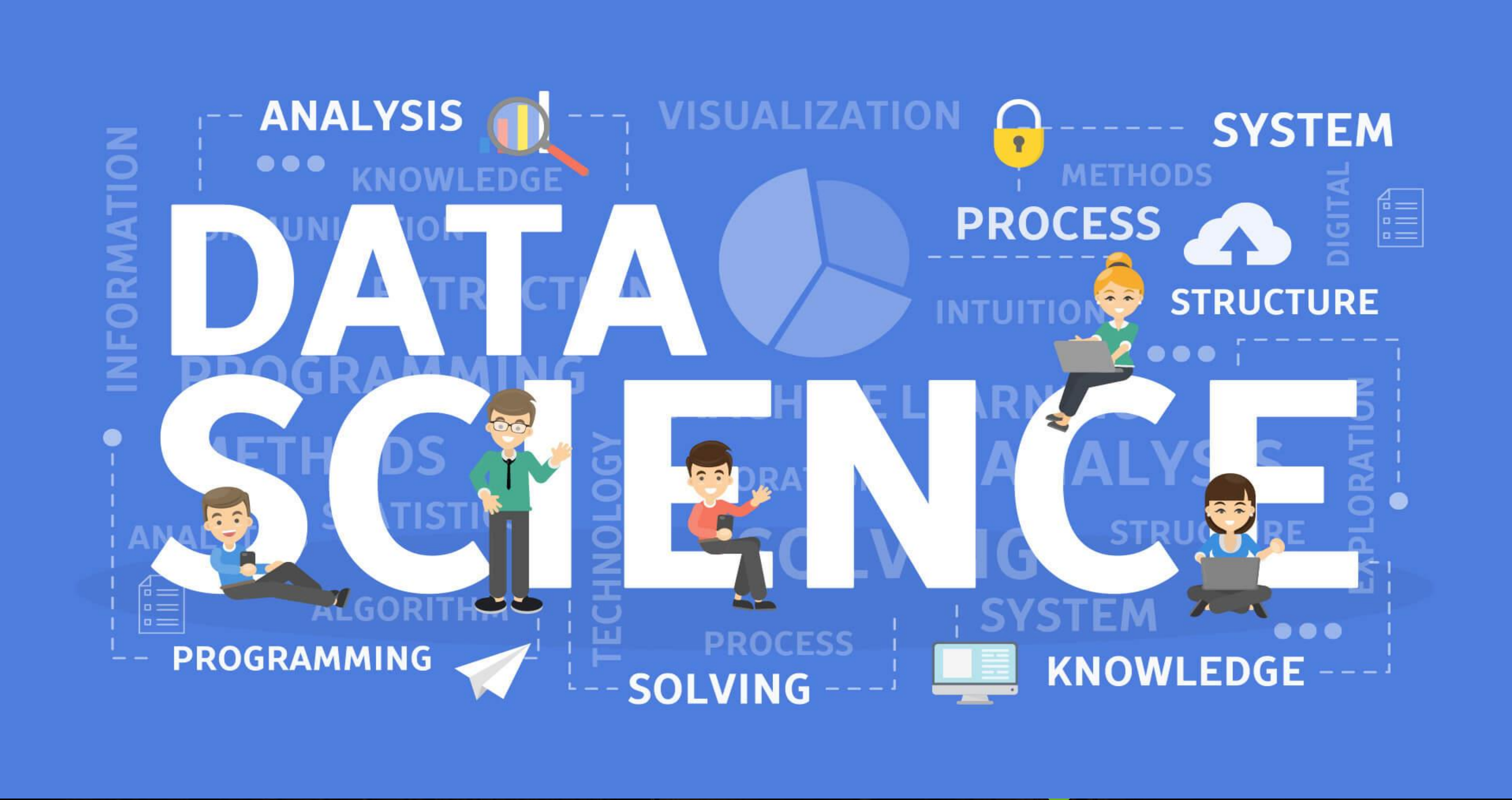

ANALYSIS
KNOWLEDGE
VISUALIZATION
SYSTEM
METHODS
PROCESS
DIGITAL
INFORMATION
DATA
SCIENCE
INTUITION
STRUCTURE
PROGRAMMING
TECHNOLOGY
SYSTEM
EXPLORATION
PROGRAMMING
PROCESS
SOLVING
KNOWLEDGE

Ex:No.#01

Download, install and explore the features of NumPy, SciPy, Jupyter, Statsmodels and Pandas packages

01 numpy

02 scipy

03 pandas

04 Statsmodels

05 jupyter

- NumPy, Pandas, SciPy: Data Analytics & Computation
- Statsmodels: Statistical Modeling & Mathematics
- Jupyter: Integrated Development Environments (IDEs) & Notebooks

Python

- We will explore the Python packages that are commonly used for data science and machine learning.
- You may need to install the packages from the terminal, Anaconda prompt, command prompt, or from the Jupyter Notebook.
- The power of Python is in the packages that are available either through the pip or conda package managers.

Procedure

Step 1: Download Python

Download Python from:
https://www.python.org/downloads/
During installation, select "Add Python to PATH".

Step 2: Install Required Packages

Open Command Prompt (Windows) or Terminal (Linux/Mac) and execute:

a. pip install numpy
b. pip install scipy
c. pip install pandas
d. pip install statsmodels
e. pip install jupyter

Or

pip install numpy scipy pandas statsmodels jupyter

Step 3: Explore the Packages

Step 2. Check Whether all the Python Libraries were installed properly.

❖ List all the Libraries Installed

python –m pip list

❖ Show detail about the Libraries Installed

python –m pip show numpy

```
Administrator: Command Prompt
C:\Users\Administrator>
C:\Users\Administrator>python -m pip show numpy
Name: numpy
Version: 2.4.0
Summary: Fundamental package for array computing in Python
Home-page: https://numpy.org
Author: Travis E. Oliphant et al.
Author-email:
License-Expression: BSD-3-Clause AND 0BSD AND MIT AND Zlib AND CC0-
1.0
Location: C:\Users\Administrator\AppData\Local\Python\pythoncore-3.
14-64\Lib\site-packages
Requires:
Required-by: pandas, patsy, scipy, statsmodels

C:\Users\Administrator>_
```

❖ List installed packages in a format suitable for sharing

python –m pip freeze

❖ Check if a library is installed from Python

❖ Run python in command mode

❖ import numpy

print(numpy.__version__)

❖ import pandas

print(pandas.__version__)

❖ import scipy

print(scipy.__version__)

❖ import statsmodels

print(statsmodels.__version__)

Step 2.a. Install numpy(Numerical Python)

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python --version
Python 3.14.2

C:\Users\Administrator>pip install numpy
'pip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>
```

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python --version
Python 3.14.2

C:\Users\Administrator>pip
'pip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>python -m pip --version
pip 25.3 from C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)

C:\Users\Administrator>
```

python -m pip --version

Administrator: Command Prompt - python -m pip install numpy

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python --version
Python 3.14.2

C:\Users\Administrator>pip
'pip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>python -m pip --version
pip 25.3 from C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)

C:\Users\Administrator>python -m pip install numpy
Requirement already satisfied: numpy in c:\users\administrator\appdata\local\python\pythoncore-3.14-64\lib\site-packages
 (2.4.0)
```

Administrator: Command Prompt

```
C:\Users\Administrator>pip
'pip' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>python -m pip --version
pip 25.3 from C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\pip (python 3.14)

C:\Users\Administrator>python -m pip install numpy
Requirement already satisfied: numpy in c:\users\administrator\appdata\local\python\pythoncore-3.14-64\lib\site-packages
 (2.4.0)

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgr
ade pip

C:\Users\Administrator> C:\Users\Administrator\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgra
de pip
Requirement already satisfied: pip in c:\users\administrator\appdata\local\python\pythoncore-3.14-64\lib\site-packages (
25.3)
Collecting pip
  Downloading pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.1.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 6.5 MB/s  0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.3
    Uninstalling pip-25.3:
      Successfully uninstalled pip-25.3
  WARNING: The scripts pip.exe, pip3.14.exe and pip3.exe are installed in 'C:\Users\Administrator\AppData\Local\Python\p
ythoncore-3.14-64\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-26.1.2

C:\Users\Administrator>
```

Step 2.a. Exploring numpy(Numerical Python)

Standard deviation

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])
print("Array:", arr)
print("Mean:", np.mean(arr))
print("Sum:", np.sum(arr))
```

Random Number Generation

```
import numpy as np

print(np.random.randint(1, 10, 5))
```

Square root

```
import numpy as np

a = np.array([1, 4, 9, 16])

print(np.sqrt(a))
```

Array Reshaping

```
import numpy as np

a = np.arange(12)

print(a.reshape(3, 4))
```

Step 2.a. Exploring numpy(Numerical Python) To Do

To Do

Feature	Description
Multidimensional Arrays	Supports 1D, 2D, and n-dimensional arrays
High Performance	Faster than Python lists for numerical operations
Mathematical Functions	Built-in arithmetic and scientific functions
Broadcasting	Operates on arrays of different shapes automatically
Indexing & Slicing	Efficient data access and modification
Statistical Functions	Mean, median, variance, standard deviation, etc.
Linear Algebra	Matrix operations, determinants, inverses, eigenvalues
Random Module	Generates random numbers for simulations
Array Reshaping	Changes array dimensions without copying data

Step 2.b. Install scipy(Scientific python)

Select Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python -m pip install scipy
```

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python -m pip install scipy
Collecting scipy
  Downloading scipy-1.18.0-cp314-cp314-win_amd64.whl.metadata (61 kB)
Requirement already satisfied: numpy<2.8,>=2.0.0 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from scipy) (2.4.0)
Downloading scipy-1.18.0-cp314-cp314-win_amd64.whl (37.3 MB)
   ---------------------------------------- 37.3/37.3 MB 4.9 MB/s  0:00:07
Installing collected packages: scipy
Successfully installed scipy-1.18.0

C:\Users\Administrator>
```

Step 2.b. Exploring scipy(Scientific python)

Statistics Analysis

```
from scipy import stats

data = [5, 10, 15, 20, 25]

print("Mean:", stats.tmean(data))
print("Variance:", stats.tvar(data))
```

Image Processing

```
from scipy import ndimage
import numpy as np

image = np.array([[1,2],[3,4]])
rotated = ndimage.rotate(image, 45)
print(rotated)
```

Fourier Transform

```
from scipy.fft import fft

data = [1, 2, 3, 4]
print(fft(data))
```

Sparse Matrix

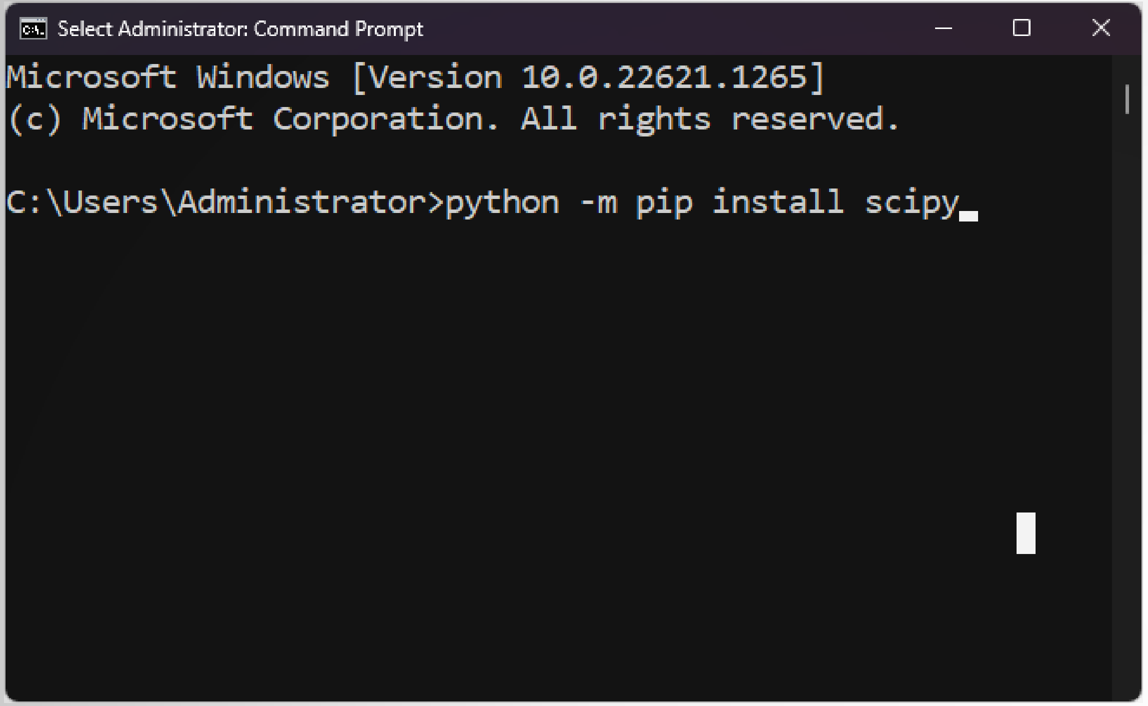
```
from scipy.sparse import csr_matrix

matrix = csr_matrix([[0,0,1],[1,0,0]])
print(matrix)
```

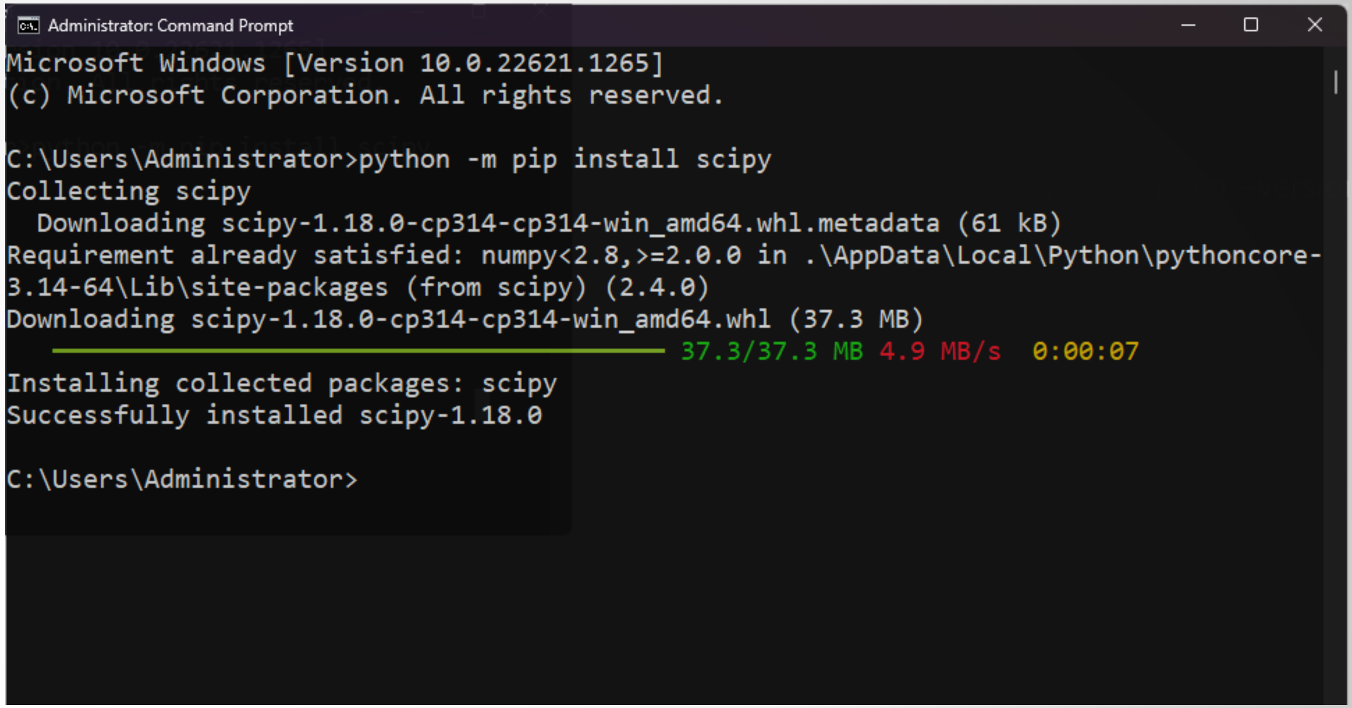
Step 2.b. Exploring scipy(Scientific python) To Do

To Do

Feature	Description
Scientific Computing	Advanced mathematical and engineering computations
Numerical Integration	Computes integrals and solves differential equations
Optimization	Finds optimal solutions for mathematical problems
Statistics	Performs descriptive and inferential statistical analysis
Linear Algebra	Matrix operations, determinants, inverses, eigenvalues
Signal Processing	Digital signal filtering and analysis
Image Processing	Image transformations and filtering
Interpolation	Estimates intermediate values from known data
Fourier Transform	Frequency-domain analysis using FFT
Sparse Matrices	Efficient storage and computation for sparse data

Step 2.c. Install pandas (Python library used for data manipulation, analysis, and cleaning)





Step 2.b. Exploring pandas (Python library used for data manipulation, analysis, and cleaning)

```
import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)

print(df)
print(df.describe())
```

Merging and Joining Datasets

```
import pandas as pd

df1 = pd.DataFrame({"ID":[1,2], "Name":["A","B"]})
df2 = pd.DataFrame({"ID":[1,2], "Marks":[90,85]})

print(pd.merge(df1, df2, on="ID"))
```

Time Series

```
import pandas as pd

dates = pd.date_range("2026-01-01", periods=5)
print(dates)
```

Step 2.b. Exploring pandas (Python library used for data manipulation, analysis, and cleaning)

To Do

Feature	Description
Series and DataFrame	Efficient 1D and 2D data structures
Data Import/Export	Reads and writes CSV, Excel, JSON, SQL, etc.
Data Cleaning	Handles missing values and duplicates
Data Selection & Filtering	Retrieves required rows and columns
Statistical Functions	Mean, median, sum, max, min, standard deviation
Data Sorting	Sorts data in ascending or descending order
Grouping & Aggregation	Groups data and computes summaries
Merging & Joining	Combines multiple datasets
Time Series Analysis	Handles date and time-based data

Step 2.d. Install statsmodel (exploring data, conducting statistical tests, and estimating statistical models)

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python -m pip install statsmodels
Collecting statsmodels
  Downloading statsmodels-0.14.6-cp314-cp314-win_amd64.whl.metadata (9.8 kB)
Requirement already satisfied: numpy<3,>=1.22.3 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packag
es (from statsmodels) (2.4.0)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-pack
ages (from statsmodels) (1.18.0)
Requirement already satisfied: pandas!=2.1.0,>=1.4 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-pac
kages (from statsmodels) (2.3.3)
Collecting patsy>=0.5.6 (from statsmodels)
  Downloading patsy-1.0.2-py2.py3-none-any.whl.metadata (3.6 kB)
Collecting packaging>=21.3 (from statsmodels)
  Downloading packaging-26.2-py3-none-any.whl.metadata (3.5 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-
packages (from pandas!=2.1.0,>=1.4->statsmodels) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (
from pandas!=2.1.0,>=1.4->statsmodels) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages
 (from pandas!=2.1.0,>=1.4->statsmodels) (2025.3)
Requirement already satisfied: six>=1.5 in .\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages (from
 python-dateutil>=2.8.2->pandas!=2.1.0,>=1.4->statsmodels) (1.17.0)
Downloading statsmodels-0.14.6-cp314-cp314-win_amd64.whl (9.6 MB)
   ---------------------------------------- 9.6/9.6 MB 5.6 MB/s  0:00:01
Downloading packaging-26.2-py3-none-any.whl (100 kB)
Downloading patsy-1.0.2-py2.py3-none-any.whl (233 kB)
Installing collected packages: patsy, packaging, statsmodels
Successfully installed packaging-26.2 patsy-1.0.2 statsmodels-0.14.6

C:\Users\Administrator>
```

Step 2.d. Exploring statsmodel (exploring data, conducting statistical tests, and estimating statistical models)

```
import statsmodels.api as sm
import numpy as np

x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 3, 5, 4, 6])

x = sm.add_constant(x)

model = sm.OLS(y, x).fit()

print(model.summary())
```

Linear Regression

```
import statsmodels.api as sm
import numpy as np

x = np.array([1, 2, 3, 4, 5])
y = np.array([2, 4, 5, 4, 6])

x = sm.add_constant(x)      # Add intercept
model = sm.OLS(y, x).fit()

print(model.summary())
```

Step 2.d. Exploring statsmodel (exploring data, conducting statistical tests, and estimating statistical models)

To Do

Feature	Description
Linear Regression	Ordinary Least Squares (OLS) and regression models
Statistical Tests	t-test, z-test, F-test, Chi-square test
Time Series Analysis	ARIMA, SARIMA, forecasting models
ANOVA	Compares means across multiple groups
Generalized Linear Models	Logistic, Poisson, and Gamma regression
Econometric Analysis	Models for economics and finance
Formula-Based Modeling	R-style statistical formulas
Descriptive Statistics	Mean, variance, correlation, covariance
Model Diagnostics	R^2, p-values, confidence intervals, residual analysis

Step 2.e. Install jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

Administrator: Command Prompt

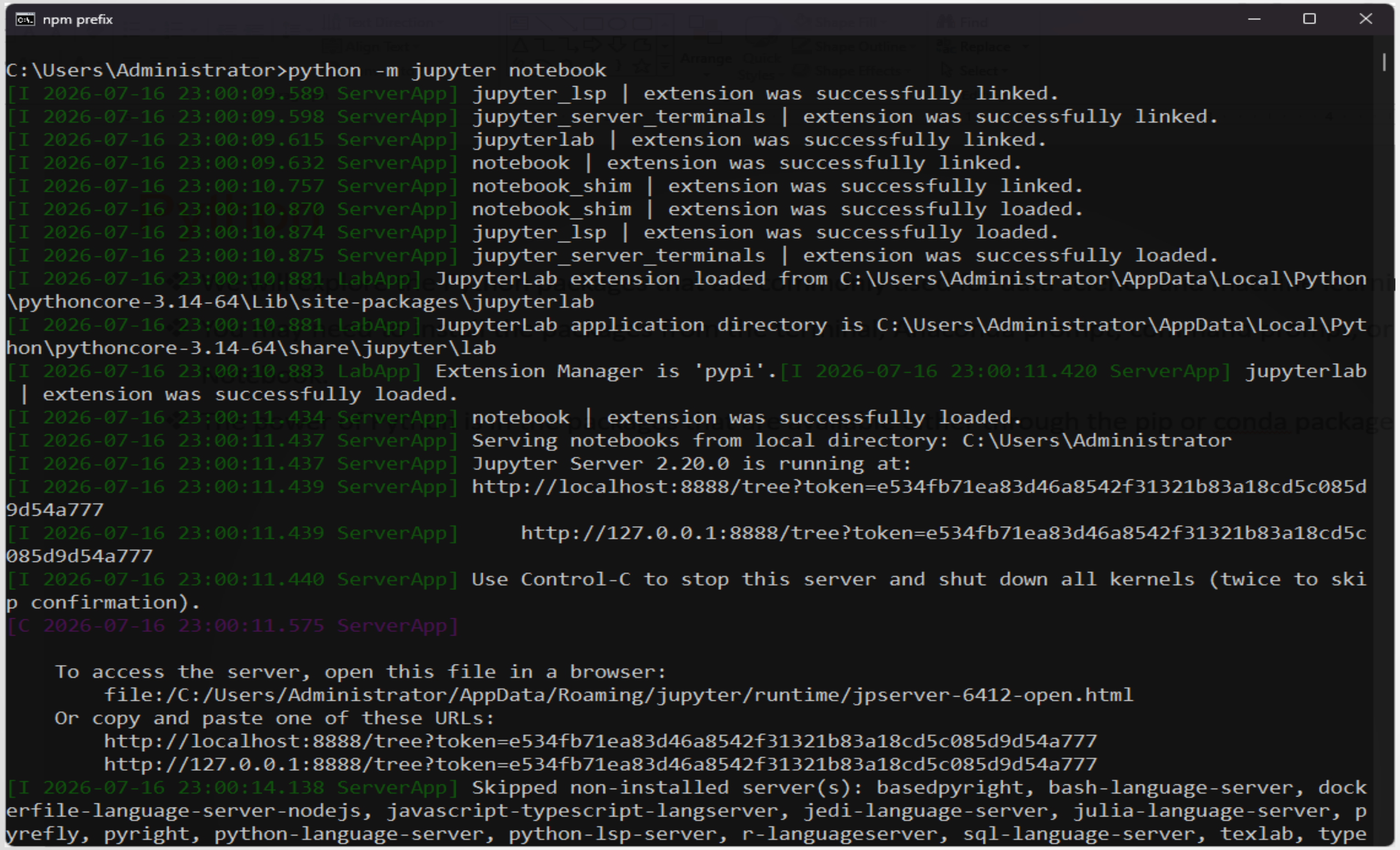
```
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>python -m pip install jupyter
Collecting jupyter
  Downloading jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)
Collecting notebook (from jupyter)
  Downloading notebook-7.6.0-py3-none-any.whl.metadata (10 kB)
Collecting jupyter-console (from jupyter)
  Downloading jupyter_console-6.6.3-py3-none-any.whl.metadata (5.8 kB)
Collecting nbconvert (from jupyter)
  Downloading nbconvert-7.17.1-py3-none-any.whl.metadata (8.4 kB)
Collecting ipykernel (from jupyter)
  Downloading ipykernel-7.3.0-py3-none-any.whl.metadata (4.5 kB)
Collecting ipywidgets (from jupyter)
  Downloading ipywidgets-8.1.8-py3-none-any.whl.metadata (2.4 kB)
Collecting jupyterlab (from jupyter)
  Downloading jupyterlab-4.6.1-py3-none-any.whl.metadata (16 kB)
Collecting comm>=0.1.1 (from ipykernel->jupyter)
```

Select Administrator: Command Prompt

```
  Consider adding this directory to PATH or, if you prefer to suppress this warning
, use --no-warn-script-location.
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 83/85 [notebook]  WARNING: The script j
upyter-notebook.exe is installed in 'C:\Users\Administrator\AppData\Local\Python\py
thoncore-3.14-64\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning
, use --no-warn-script-location.
Successfully installed anyio-4.14.2 argon2-cffi-25.1.0 argon2-cffi-bindings-25.1.0
arrow-1.4.0 asttokens-3.0.2 async-lru-2.3.0 attrs-26.1.0 babel-2.18.0 beautifulsoup
4-4.15.0 bleach-6.4.0 cffi-2.1.0 comm-0.2.3 debugpy-1.8.21 decorator-5.3.1 defusedx
ml-0.7.1 executing-2.2.1 fastjsonschema-2.21.2 fqdn-1.5.1 h11-0.16.0 httpcore-1.0.9
 httpx-0.28.1 ipykernel-7.3.0 ipython-9.15.0 ipython-pygments-lexers-1.1.1 ipywidge
ts-8.1.8 isoduration-20.11.0 jedi-0.20.0 json5-0.15.0 jsonpointer-3.1.1 jsonschema-
4.26.0 jsonschema-specifications-2025.9.1 jupyter-1.1.1 jupyter-builder-1.1.0 jupyt
er-client-8.9.1 jupyter-console-6.6.3 jupyter-core-5.9.1 jupyter-events-0.12.1 jupy
ter-lsp-2.3.1 jupyter-server-2.20.0 jupyter-server-terminals-0.5.4 jupyterlab-4.6.1
 jupyterlab-pygments-0.3.0 jupyterlab-server-2.28.0 jupyterlab_widgets-3.0.16 lark-
1.3.1 matplotlib-inline-0.2.2 mistune-3.3.3 nbclient-0.11.0 nbconvert-7.17.1 nbform
at-5.10.4 nest-asyncio2-1.7.2 notebook-7.6.0 notebook-shim-0.2.4 pandocfilters-1.5.
1 parso-0.8.7 platformdirs-4.10.0 prometheus-client-0.25.0 prompt_toolkit-3.0.52 ps
util-7.2.2 pure-eval-0.2.3 pycparser-3.0 pygments-2.20.0 python-json-logger-4.1.0 p
ywinpty-3.0.5 pyyaml-6.0.3 pyzmq-27.1.0 referencing-0.37.0 rfc3339-validator-0.1.4
rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.6.3 send2trash-2.1.0 soup
sieve-2.8.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.5.7 traitlet
s-5.15.1 typing-extensions-4.16.0 uri-template-1.3.0 wcwidth-0.8.2 webcolors-25.10.
0 webencodings-0.5.1 websocket-client-1.9.0 widgetsnbextension-4.0.15

C:\Users\Administrator>
```

Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

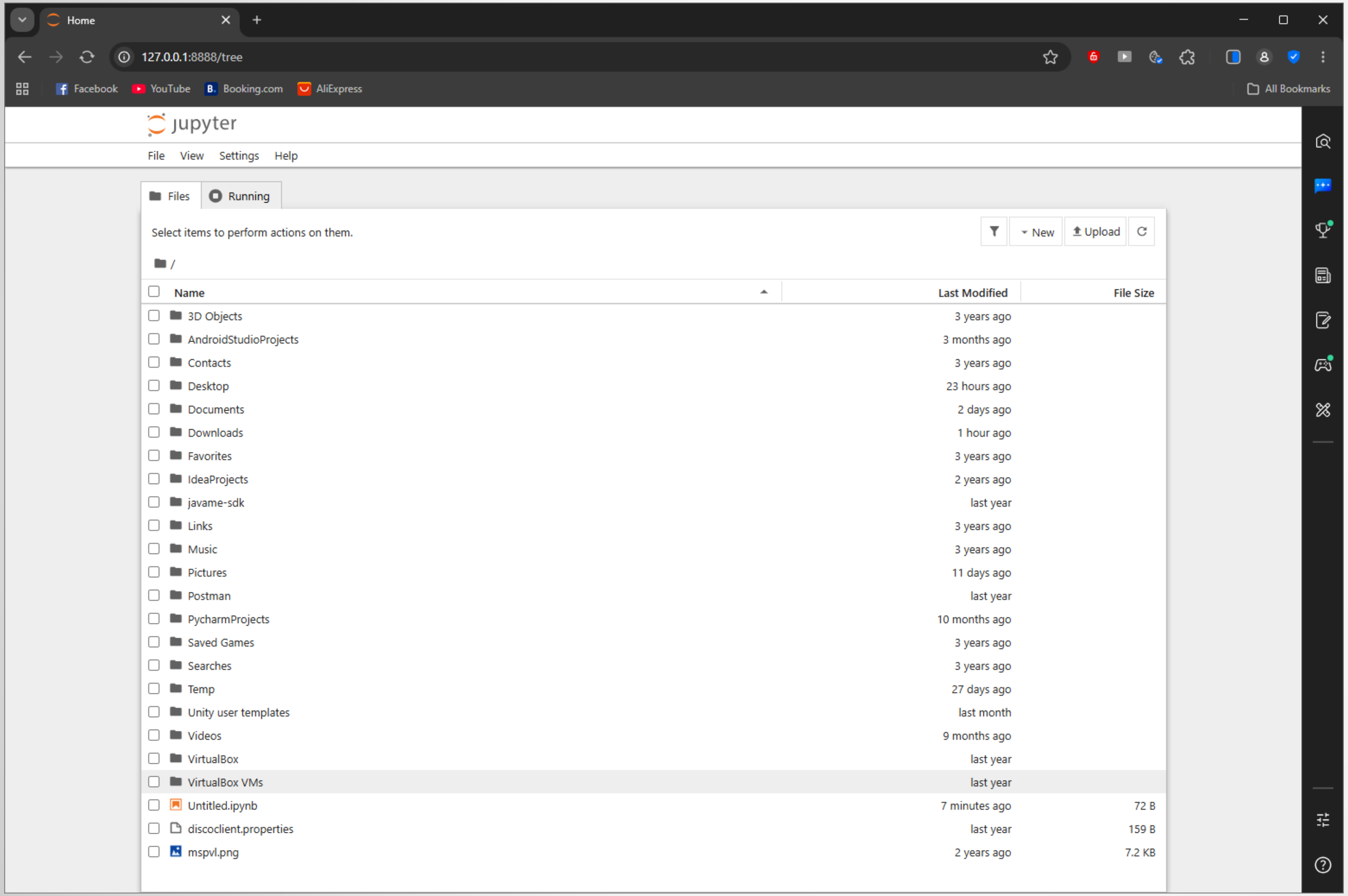
C:\Users\Administrator>python -m jupyter notebook



http://127.0.0.1:8888/tree?token=e534fb71ea83d46a8542f31321b83a18cd5c085d9d54a777

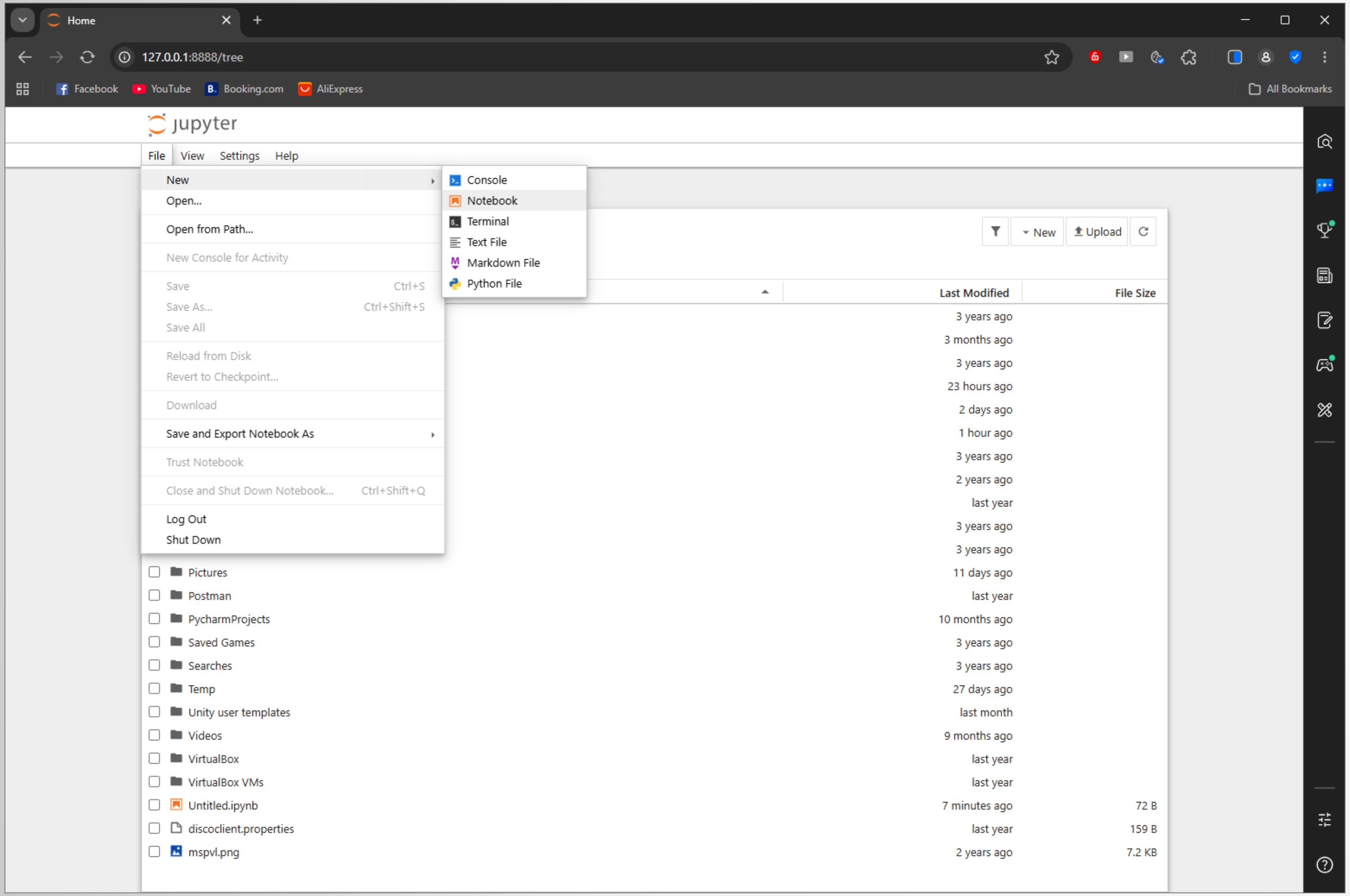
Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

http://127.0.0.1:8888/tree?token=e534fb71ea83d46a8542f31321b83a18cd5c085d9d54a777



Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

http://127.0.0.1:8888/tree?token=e534fb71ea83d46a8542f31321b83a18cd5c085d9d54a777



Create a New Notebook

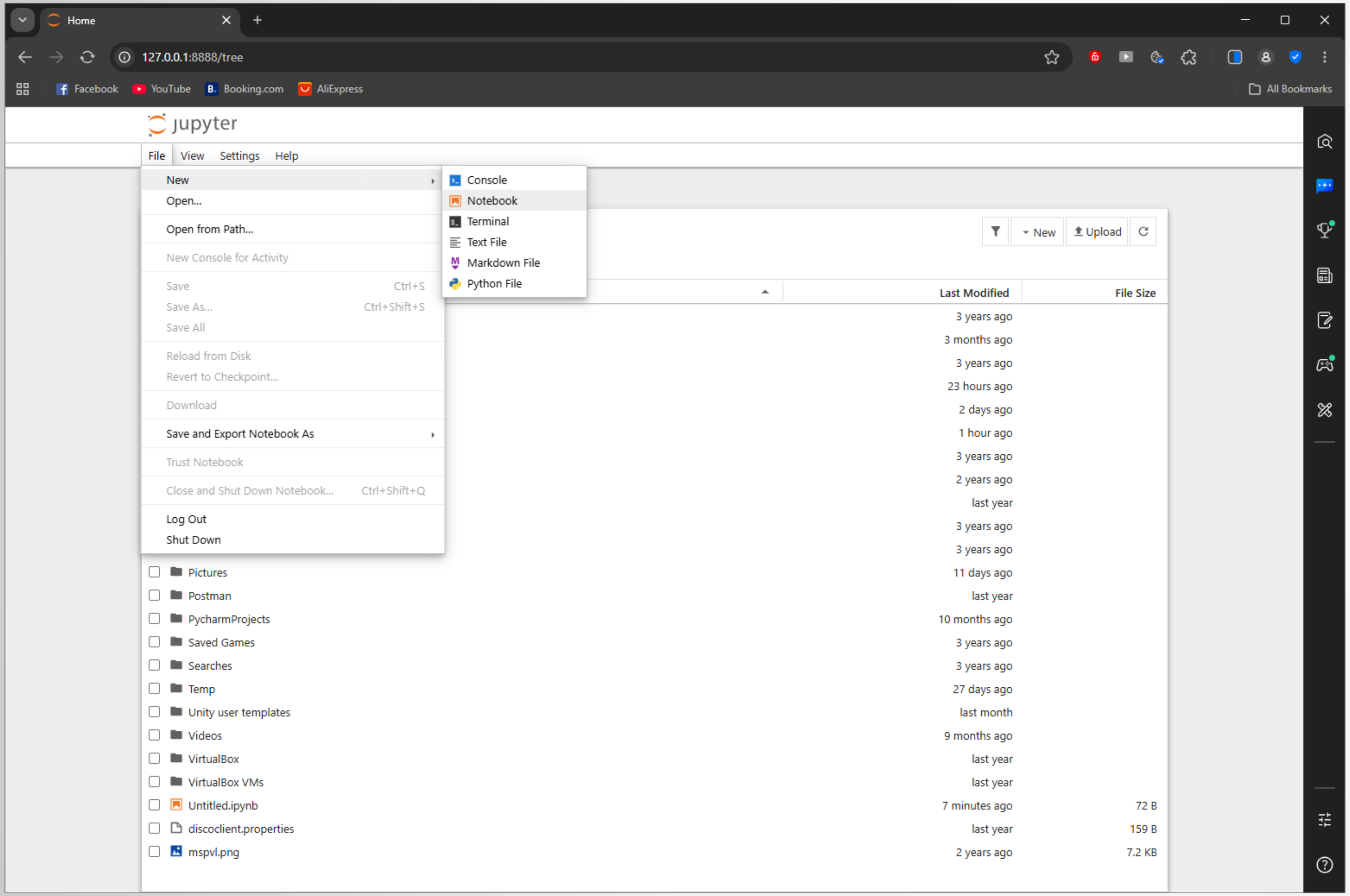
Click New → Python 3

(or the available Python kernel).

A new notebook (.ipynb) opens.

Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

http://127.0.0.1:8888/tree?token=e534fb71ea83d46a8542f31321b83a18cd5c085d9d54a777



Test Jupyter with a Simple Program

In the first cell, type:

print("Welcome to Jupyter Notebook")

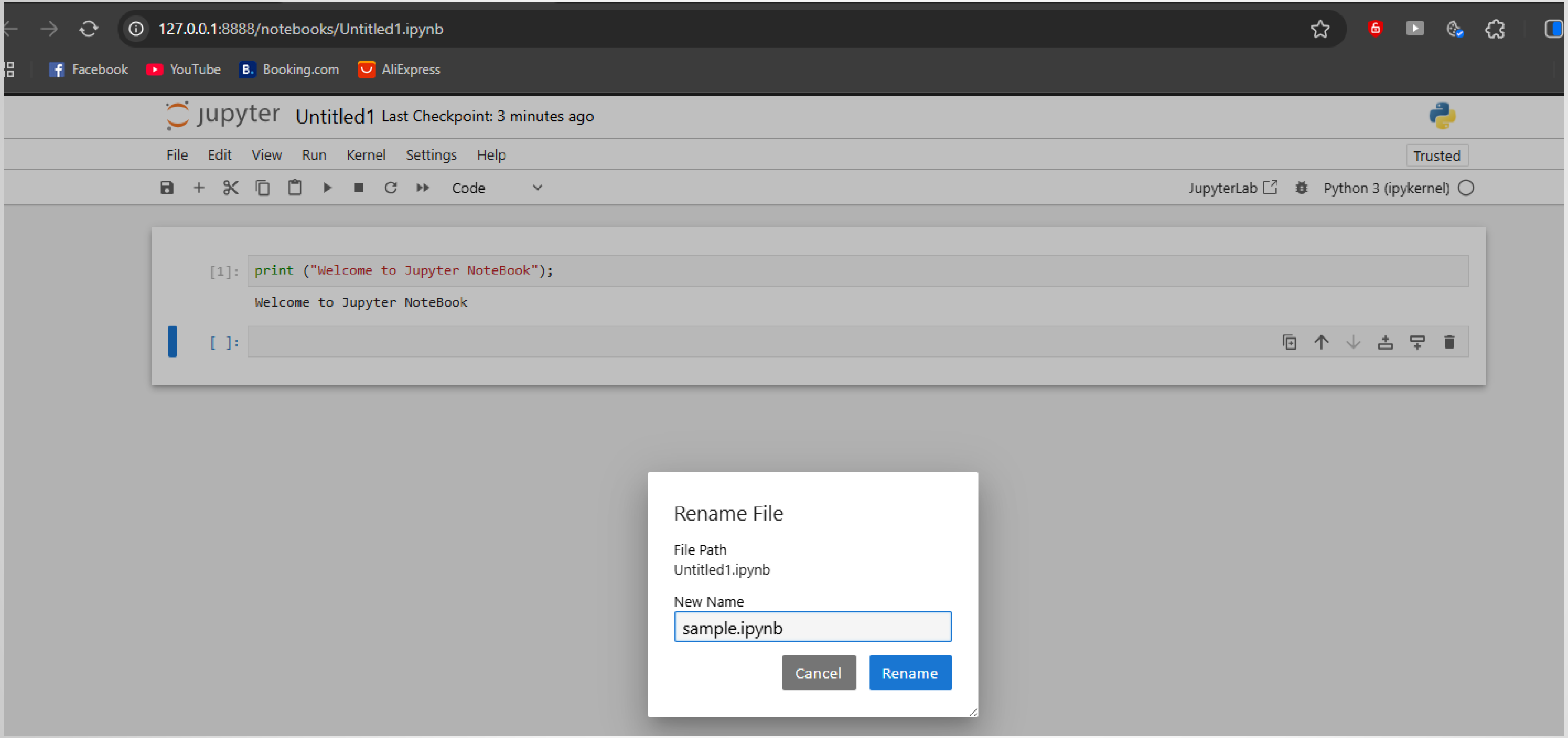
Press Shift + Enter.

Output

Welcome to Jupyter Notebook

Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

http://127.0.0.1:8888/tree?token=e534fb71ea83d46a8542f31321b83a18cd5c085d9d54a777



Step 2.e. Exploring jupyter (interactive documents combining live code, narrative text, equations, and visual charts)

To Do

Feature	Description
Interactive Code Execution	Runs code cell by cell with immediate output
Multi-language Support	Supports Python, R, Julia, Scala, etc.
Markdown Support	Creates formatted text and documentation
Data Visualization	Displays charts and graphs inline
Data Analysis	Integrates with NumPy, Pandas, SciPy, etc.
Multiple Cell Types	Code, Markdown, and Raw cells
Inline Output	Shows results, tables, and plots below cells
Package Management	Install libraries using pip commands
Auto Save	Saves notebooks automatically as .ipynb files
Sharing & Export	Export notebooks to HTML, PDF, or Python scripts
Keyboard Shortcuts	Quick execution and notebook editing
ML Integration	Works with Scikit-learn, TensorFlow, PyTorch, and more

References

https://apmonitor.com/pds/index.php/Main/InstallPythonPackages